

Advanced approximate and symbolic computing

Tamás Vinkó

University of Szeged



Function interpolation and approximation II.

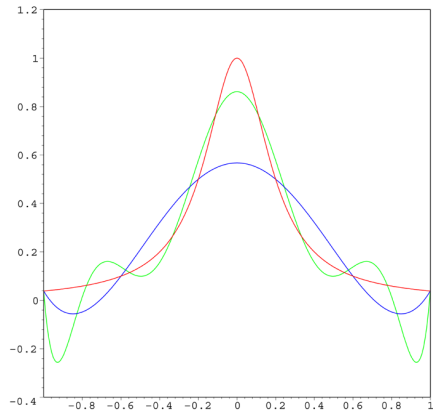
Motivation

- The Lagrange interpolation can give oscillating result in case of lots of basis points or quickly changing $f(x)$ function.
- There is a simple example, given by Runge:

$$f(x) = \frac{1}{1+x^2}$$

on the interval $[-1, 1]$, base-points $x_i = \frac{2i}{n} - 1$ ($i = 0, 1, \dots, n-1$).

- What is the Lagrange interpolation for a given n ?



red: Runge function

blue: Lagrange of degree 5

green: Lagrange of degree 9.

- We can handle this problem by
 - splitting the interval $[a, b]$ into multiple subintervals,
 - and using the base points on these subintervals we make the Lagrange polynomials (which could be of smaller degree).

- What shall we do in the joining points???

Spline interpolation

- *Spline interpolation* gives an elegant solution to the problem
- Let us introduce first some definitions.
- Suppose that the basis points are ordered, and for all x_k basis points we have the function values f_k . We work on the interval $[a, b]$ and with the splitting

$$\Delta = \{a = x_0 < x_1 < \cdots < x_n = b\}. \quad (1)$$

10.1. Definition. For a given Δ splitting and given $1 \leq m$ natural number, the spline of order- m is such function $s(x) \in C^{m-1}[a, b]$ which is at most a polynomial of at most order m on each subinterval $[x_{k-1}, x_k]$.

Given a Δ split, the set of all splines of degree m is:

$$S_{\Delta, m} \stackrel{\text{def}}{=} \{s(x) \in C^{m-1}[a, b] \quad | \quad s(x) \equiv p_k(x) \in \mathcal{P}_m \\ \text{if } x_{k-1} \leq x \leq x_k, \quad k = 1, 2, \dots, n\}$$

$C^{m-1}[a, b]$: is the set of $m - 1$ -times continuously differentiable functions.

$\mathcal{P}_m$ is the set of polynomials of order m

- In case of $m = 1$ we have *linear*, in case of $m = 2$, we have *quadratic*, and in case of $m = 3$ we have *cubic* spline.
- In the following we assume $s(x_k) = f_k$.
- We are going to discuss the details of cubic-spline interpolation.

Given Δ split, find the cubic spline

Given $1 \leq n$ and the split

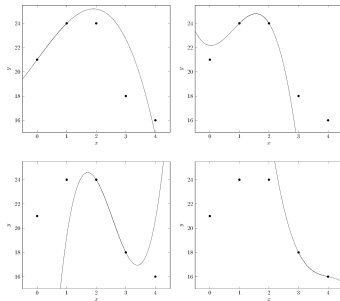
$$\Delta = \{a = x_0 < x_1 < \cdots < x_n = b\}$$

and the corresponding function values. $\{f_0, f_1, \dots, f_n\}$.

Find the cubic spline $s(x) \in C^2[a, b]$ which satisfies

$$s(x_k) = f_k \quad k = 0, 1, 2, \dots, n. \tag{2}$$

How the result will look like?



$$s(x) = \begin{cases} -0.30357x^3 + 3.3036x + 21 & \text{if } x \in [0, 1] \\ -1.4821x^3 + 3.5357x^2 - 0.23214x + 22.179 & \text{if } x \in (1, 2] \\ 3.2321x^3 - 24.750x^2 + 56.339x - 15.536 & \text{if } x \in (2, 3] \\ -1.4464x^3 + 17.357x^2 - 69.982x + 110.79 & \text{if } x \in (3, 4] \end{cases}$$

Now, we are using the white/black board to derive the formulas.

At the end of the derivation we have the following theorem (and by the derivation we also have the proof):

10.2. Proposition. *For any given Δ split, function values $f_0, f_1, \dots, f_n$ and boundary constraints f'_0, f'_n we have exactly one cubic spline $s(x) \in S_{\Delta,3}$ which satisfies the interpolation constraints as well as*

$$s'(x_0) = f'_0$$

$$s'(x_n) = f'_n$$

hold

The system of linear equations we have obtained has diagonally dominant matrix

$$|a_{jj}| > \sum_{k \neq j} |a_{kj}| \quad (j = 1, \dots, n).$$

For this kind of matrix we always have solution (i.e. the matrix is invertible).

Moreover, it is also true that

10.3. Proposition. *The coefficients of the cubic spline can be obtained using $O(n)$ operations.*

The proof of this theorem is that the system of linear equations has tridiagonal matrix.